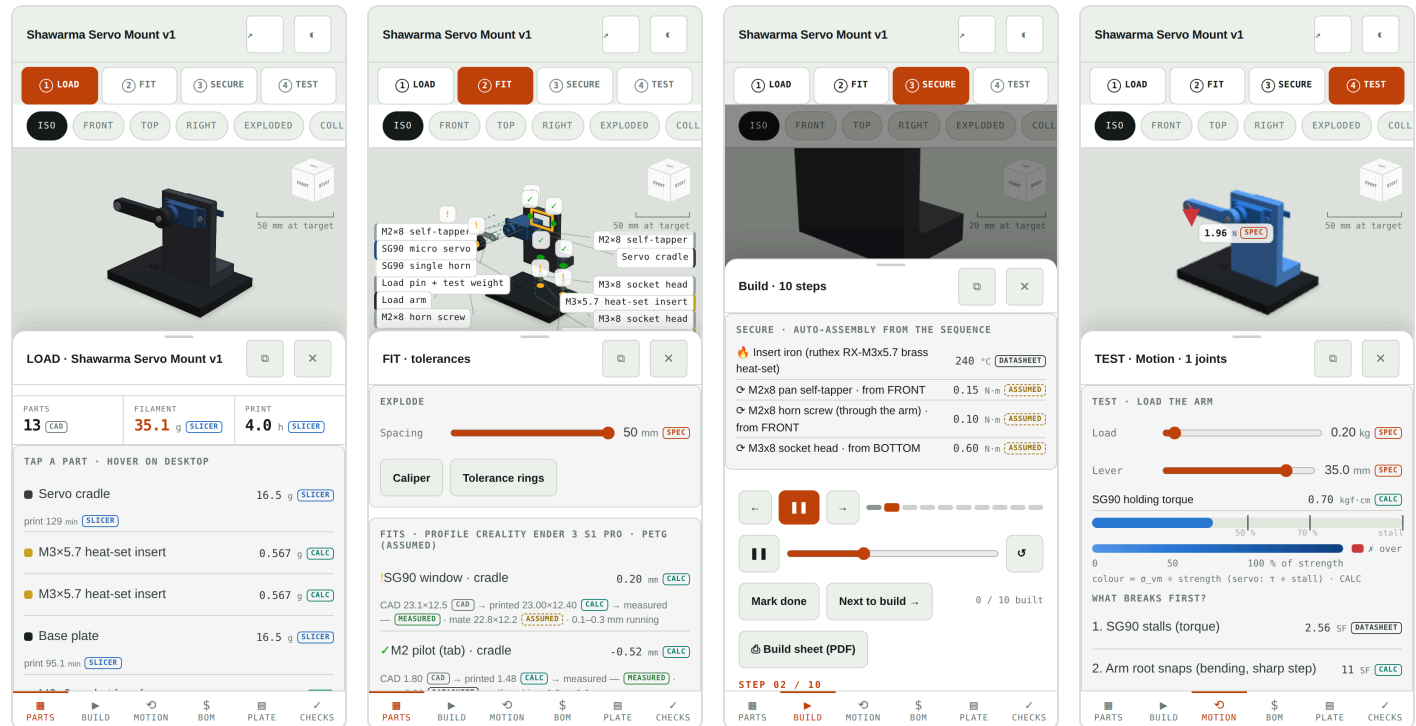


See if it fits, bolts up and holds — before you print it

One link. Spin it, explode it, measure it, load it until something gives. Every number says where it came from.



1 LOAD

mass + print time from real G-code

2 FIT

holes coloured by how your printer prints

3 SECURE

auto-build · 240 °C inserts · screw sides

4 TEST

drag the load · what breaks first

0.51 kg

SG90 stalls at 35 mm **CALC**

2.24 kg

arm root snaps, sharp step **CALC**

4.05 kg

with a 2 mm fillet **CALC**

0.10 %

PyBullet vs hand calc

Caught before a gram was printed

- **Tips over:** CoG 8.0 mm outside the base with 200 g → base under the load, now 12.1 mm inside.
- **Pilot too tight:** Ø1.5 prints Ø1.13 → Ø1.8 in CAD prints Ø1.48, a clean M2 bite.
- **SG90 may not fit:** datasheets disagree by 0.8 mm → worst case -0.10 mm. Calipers first.

Specs

- Ender 3 S1 Pro · PETG · 35.2 g · 4 h print (PrusaSlicer G-code)
- Unit #1 out of pocket CAD \$2.28 (hardware on the shelf)
- One self-contained HTML · works offline · shares as a link

Every number has a source

- MEASURED · CALC · CAD · SLICER · DATASHEET · ASSUMED · SPEC
- Any ASSUMED input → the check reads UNVERIFIED, never PASS
- Verified: 9 gate families, every legacy gate, 390 px phone, dark + light

Get one

- \$450 — LOAD + FIT + SECURE
- \$1,200 — + TEST, checks, measurement list
- First one free for a testimonial